Weekly Diary – Week 18

Internship Organization: Vikaspedia
Duration: 05/05/2026 – 10/05/2026

Objective of the Week

The objective of this week was to implement broken-link detection rules, design auto-replacement mechanisms, and integrate intelligent replacement strategies for suggesting alternative working URLs automatically.

Date- 05/May

Studied different criteria used for identifying broken links in web crawling systems. Implemented validation logic to classify URLs as broken when response codes were greater than or equal to 400, or when DNS failures and timeout exceptions occurred. Also added support for detecting inaccessible or dead domains. Conducted tests with various categories of broken links to verify whether the crawler correctly identified failed URLs under different scenarios.

Date- 06/May

Researched multiple strategies for implementing automatic replacement of broken links. Analyzed predefined mapping techniques where old URLs are directly mapped to new working URLs. Also studied pattern-based replacement strategies that generate replacement URLs automatically based on similar structures or keywords. Prepared the overall workflow for integrating replacement functionality into the crawler system.

Day- 07/May

Implemented predefined mapping-based replacement logic in the backend system. Developed functionality where broken URLs are automatically checked against stored mapping records to locate corresponding replacement links. Created sample mapping datasets for testing purposes and verified whether the crawler could correctly suggest replacement URLs whenever matching mappings were available.

Date- 08/May

Developed pattern-based replacement functionality for handling URLs that follow predictable naming structures. Implemented logic to extract keywords and path segments from broken URLs and generate probable replacement suggestions automatically. Tested the system using sample article and blog URLs to analyze the accuracy of generated replacements. Also documented cases where pattern-based replacement may produce incorrect or incomplete results.

Date- 09/May

Implemented search-based replacement logic for suggesting alternative URLs when predefined mappings are unavailable. Developed functionality to extract keywords from broken URLs and compare them with internally stored content references to identify similar pages. Conducted testing with sample datasets and evaluated the relevance of suggested replacement URLs generated by the system.

Date- 10/May

Integrated all replacement mechanisms into a unified replacement workflow. Configured the crawler to attempt predefined mapping first, followed by pattern-based and search-based approaches if no direct replacement is found. Added confidence scores and logging functionality for replacement suggestions to support future manual review. Conducted multiple tests to evaluate the accuracy and consistency of the complete replacement system.

Conclusion

By the end of this week, the system was capable of detecting broken links accurately and generating replacement suggestions using multiple intelligent replacement strategies. The crawler successfully combined validation, replacement, and logging mechanisms into a single workflow.